



"Please turn left into the alley where four kidnappers lie waiting!"

phone's already been compromised by means of the IMSI catcher which can listen in on your conversations and intercept all data. But if you were to shut off your 2G/3G/LTE data connection and switch to Wi-Fi, then the attacker has an all new problem to deal with. Worry not, it won't be long before that too is compromised.

Here's what he'll likely do. Every consumer router has backdoors built into it at the hardware level. This might seem like the beginning of a conspiracy theory but there have been numerous occasions when such backdoors were widely talked about in the mainstream media. The most recent would have been in January, where the infamous TCP-32764 vulnerability was all over the Internet. Basically, sending an encoded packet at a router with the above vulnerability would cause it to give admin access to an attacker. But this involves getting in between the router and the ISP.

There are other ways to obtain access. Let's take a look at a simple method. This involves having a computer with any linux distro designed for penetration testing (eg. Kali). You need Aircrack and oclHashcat installed. Lastly, you'll need to buy a Wi-Fi dongle that is compatible with Aircrack. When you are in the vicinity of the target, simply fire up Aircrack and start monitoring the Wi-Fi in the area. With a few seconds every station, including the ones that have their SSID hidden, will be logged. The attacker now picks out each SSID one by one till

the target router is found. Another simple way is to send an email to the to-be-victim and a simple response gives the attacker the route that the email took. The IP obtained can then be used to perform a port scan using Nmap to look for a way into the network.

Wi-Fi connections perform an action that's known as a handshake, which is a

way for devices to identify themselves. This handshake involves sending a data packet with the Wi-Fi password in the header. This packet is obtained using Aircrack and is encrypted. However, using oclHashcat or its more specialised variants, the attacker can easily start a brute force attack. We keep reading about all these major hacks with millions of passwords being stolen. These aren't actual passwords but encrypted hashes and a good portion of these hashes are decrypted to obtain the actual passwords which in turn are added to online word lists. This makes these wordlists very accurate and passwords are cracked all the more easily. By now, the attacker has obtained access to your phone, your home security system, your smart car and your Wi-Fi.

Now the attacker has multiple paths to infect your system so he gets onto the task at hand. The first step is to get backdoor access to your computers. A simple Google search for "Zero-day attacks" gets you plenty of lists where vulnerabilities present in commonly used software are disclosed with ample detail. Zero-day attacks are, by definition, so recent that security companies

BUT IF IT'S LEFT LYING AROUND?

We're assuming you wouldn't leave your devices lying around unattended because that's pretty much akin to hammering nails into your own casket. Here is a simple keylogger for instance - Ardamax - one of the best commercial keyloggers available. Now normally an antivirus program will have heuristics algorithms which will be triggered if any software tries to log every keystroke or attempt sending files or text to an unknown or random IP address. But you forget the golden words... physical access. That felt much more dramatic in our heads but let's try to outline what all can be done. Apart from a keylogger, which by the way can be programmed to send out weekly / daily reports for certain catch phrases such as userid / password or it might simply be triggered whenever an HTTPS page is accessed (read banks, email services, etc.), there are other spying tools that can be installed and even hidden from your task manager. Your antivirus too can be made to give all of these software an exception rule. Software that'll take screen shots periodically, software that'll stream video from your laptop's webcam, etc. So in effect every website visited, every credit card transaction, every user id and password combination and every action performed just got uploaded to an attackers botnet. Even your social media image can be altered. You'll suddenly find your Facebook profile liking pages that immediately raise flags that every security agency is looking at. For the NSA, you could go from being a soccer mom to a cold-blooded ISIS activist trying to recruit innocent kids on the Internet. Your user ids and password could be offered up for sale to terrorists or much worse